# Scheme and Answer Key for

## Question Paper Code : 25900564

## MAHATMA GANDHI UNIVERSITY, KOTTAYAM

### MGU-UGP (HONOURS) Regular EXAMINATION October 2025

**Third SEMESTER**

**Discipline Specific Elective (DSE) - MG3DSEPHY207 - PROPERTIES OF MATTER**

**(2024 ADMISSION ONWARDS)**

**Duration: 1 Hour 30 Minutes** **Maximum Marks: 50**

**Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

**Part A**

**Short Answer Type Questions**

**Answer any 10 questions**

**Each question carries 2 marks**

1. If a cylinder with radius (R) and length (L) is given, give the expression to find out the twisting couple of the cylinder.
   **Answer:**
   Twisting couple, $C = (\pi \eta R^4 \theta) / (2L)$. (1 mark)
   $\eta$ – rigidity modulus, R – radius, L – length, $\theta$ – angle of twist. (1 mark)
   [A] / [CO2]

2. Define the terms plane of bending and axis of bending.
   **Answer:**
   Plane of bending – the plane in which the neutral axis lies and bending occurs. (1 mark)
   Axis of bending – the line about which bending moment acts. (1 mark)
   [U] / [CO1, CO2]

3. State the relation connecting Young's modulus, Bulk modulus and Poisson's ratio.
   **Answer:**
   $E = 3K(1 – 2\sigma)$. (1 mark)
   E – Young's modulus, K – Bulk modulus, $\sigma$ – Poisson's ratio. (1 mark)
   [U] / [CO1, CO2]

4. How can we define piezoelectricity?
   **Answer:**
   Piezoelectricity is the property of certain crystals to develop an electric potential when mechanical stress is applied. (1 mark)

Example: Quartz or Rochelle salt. (1 mark)
[K] / [CO2]

5. What you mean by a cantilever?
**Answer:**
A cantilever is a beam fixed at one end and free at the other. (1 mark)
It bends when a load is applied at the free end. (1 mark)
[U] / [CO1, CO2]

6. State the assumptions made in deriving Bernoulli's principle.
**Answer:**
(i) The flow of liquid is steady and incompressible. (1 mark)
(ii) There is no loss of energy due to viscosity or friction. (1 mark)
[U] / [CO3]

7. What is meant by excess of pressure? What is its SI unit?
**Answer:**
Excess of pressure is the difference between the pressure inside and outside a curved surface. (1 mark)
SI unit: Pascal (Pa). (1 mark)
[U] / [CO3]

8. How does the viscosity of a fluid affect its terminal velocity?
**Answer:**
Viscosity increases the resistive force acting on a body moving through the fluid. (1 mark)
Hence, higher viscosity reduces terminal velocity. (1 mark)
[U] / [CO3]

9. What is meant by surface energy?
**Answer:**
Surface energy is the potential energy per unit area possessed by a liquid surface. (1 mark)
It arises due to surface tension; unit: $J/m^2$. (1 mark)
[K] / [CO3]

10. Define time period of a wave.
**Answer:**
The time period (T) is the time taken for one complete oscillation. (1 mark)
It is related to frequency by $T = 1/f$. (1 mark)
[U] / [CO4]

11. State any four properties of ultrasonic waves.
**Answer:**
(i) Frequency greater than 20 kHz. (0.5 mark)
(ii) Short wavelength. (0.5 mark)
(iii) Highly directional. (0.5 mark)
(iv) Used in SONAR, cleaning, welding etc. (0.5 mark)
[U] / [CO6]

12. Write the differential equation for a one-dimensional wave travelling along the positive x-direction.
**Answer:**

$\partial^2 y/\partial x^2 = (1/v^2)(\partial^2 y/\partial t^2)$. (1 mark)
This represents a progressive wave moving in the +x direction with velocity v. (1 mark)
[U] / [CO4]

13. What are beats?
**Answer:**
Beats are periodic variations in sound intensity. (1 mark)
They occur due to interference of two waves of slightly different frequencies. (1 mark)
[U] / [CO5]

14. Define a longitudinal wave. Give two examples.
**Answer:**
A longitudinal wave is one in which particles vibrate parallel to the direction of propagation. (1 mark)
Examples: Sound waves in air, compression waves in springs. (1 mark)
[U] / [CO5]

## Part B

### Short Essay Type Questions

**Answer any 6 questions**

**Each question carries 5 marks**

**15. Explain about the different elastic behavior of solids.**
**Answer:**
Solids obey Hooke's law for small deformations. (1 mark)
Beyond the elastic limit, permanent deformation starts. (1 mark)
Ductile materials show large plastic strain before breaking. (1 mark)
Brittle materials break without much deformation. (1 mark)
Area under stress–strain curve shows resilience and toughness. (1 mark)
[U] / [CO1]

**16. How can we compare moments of inertia using torsion pendulum?**
**Answer:**
Time period $T = 2\pi\sqrt{(I/C)}$. (1 mark)
Measure time period of platform alone ($T_0$) and with the body (T). (1 mark)
Moment of inertia of body, $I = (C/4\pi^2)(T^2 - T_0^2)$. (1 mark)
For two bodies, $I_1/I_2 = (T_1^2 - T_0^2)/(T_2^2 - T_0^2)$. (1 mark)
Hence moments of inertia can be compared experimentally. (1 mark)
[K] / [CO2]

**17. Draw and explain the stress–strain diagram.**
**Answer:**
In the beginning, stress is directly proportional to strain (Hooke's law). (1 mark)
Elastic limit is the point up to which the body regains its shape. (1 mark)
Beyond yield point, plastic deformation begins. (1 mark)
Maximum stress is the ultimate strength; then necking and fracture occur. (1 mark)
Curve differs for ductile and brittle materials. (1 mark)
[An] / [CO1, CO2]

**18. Explain the molecular theory of surface tension.**
**Answer:**
Molecules inside the liquid attract equally in all directions. (1 mark)
Surface molecules have a net inward force. (1 mark)
Surface acts like a stretched elastic membrane. (1 mark)
Surface tension is the force per unit length. (1 mark)
It causes spherical drops and capillary rise. (1 mark)
[U] / [CO3]

**19. Calculate the Reynold's number for water flowing through a pipe of diameter 0.1 m at a velocity 2 m/s. Given density = 1000 kg/m³ and viscosity = 0.001 Pa·s.**
**Answer:**
Formula: $Re = (\rho v D) / \mu$. (1 mark)
Substitute: $Re = (1000 \times 2 \times 0.1) / 0.001$. (1 mark)
$Re = 2 \times 10^5$. (1 mark)
$Re > 4000$, so the flow is turbulent. (1 mark)
Reynolds number is dimensionless. (1 mark)
[A] / [CO3]

**20. How is ultrasonic used in flaw detection?**
**Answer:**
Ultrasonic waves reflect from cracks or defects in metals. (1 mark)
A pulse is sent and reflected echoes are received. (1 mark)
Time of echo gives the position of the flaw. (1 mark)
No echo indicates a good region. (1 mark)
Used for non-destructive testing in industries. (1 mark)
[U] / [CO6]

**21. Calculate the velocity with which a source of frequency 10000 Hz should approach an observer at rest in order to produce a Doppler shift of 200 Hz. Velocity of sound = 332 m/s.**
**Answer:**
Given $f = 10000$ Hz, $f' = 10200$ Hz, $v = 332$ m/s. (1 mark)
Formula: $f' = f \times [v / (v - v_s)]$. (1 mark)
$10200/10000 = 332 / (332 - v_s)$. (1 mark)
$v_s = 6.5$ m/s. (1 mark)
Hence, the source must approach at about 6.5 m/s. (1 mark)
[U] / [CO6]

**22. Explain the significance of Mach number in classifying the motion of an aircraft.**
**Answer:**
Mach number $M = V / a$ (speed / speed of sound). (1 mark)
$M < 1$: subsonic, $M = 1$: sonic, $M > 1$: supersonic. (1 mark)
Compressibility effects increase with M. (1 mark)
Shock waves form when $M \geq 1$. (1 mark)
Mach number helps classify aircraft speed and design. (1 mark)
[An] / [CO5]